

# DANYEL YAMANOGULLU

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## PROFILE

Control and Automation Engineer specializing in the convergence of classical Control Theory and modern Generative AI architectures. I build intelligent systems that bridge hardware control logic with autonomous AI, leveraging a strong foundation in state estimation and stochastic modeling to drive scalable, data-centric workflows.

Core Focus Areas:

- Architecting robust RAG pipelines and LLM orchestration frameworks.
- Training and optimizing ML models (XGBoost, Neural Networks) for predictive analytics.
- Engineering data ingestion pipelines to transform raw hardware/sensor metrics into actionable AI features.
- Developing high-load, production-ready backends using Python, FastAPI, and Docker.

## EXPERIENCES

- 03/2026 – 06/2026 **Engineering Intern | ENKO**
- Evaluated PLC architectures and mapped sensor data flow across industrial protocols (e.g., Modbus, MQTT).
  - Engineered telemetry data ingestion pipelines for predictive maintenance models.
- 06/2025 – 09/2025 **Engineering Intern | ADATECH**
- Developed an HR analytics dashboard featuring Optimized Team and Risk Analysis modules.
  - Built an XGBoost-based predictive model for early detection of turnover risks with 91.7% accuracy.
  - Designed data-driven workflows, improving workforce allocation efficiency by 15%.

## EDUCATION

- 2020 – 2026 **Control and Automation Engineering (GPA: 3.01 / 4.00)**  
Yıldız Technical University
- 2016 – 2020 **Fener Rum High School**

## SKILLS

- AI/ML:** PyTorch, XGBoost, LangChain
- Infrastructure:** ROS, Docker, Kubernetes, CI/CD, FastAPI, Simulink
- Languages:** Python, C/C++, MATLAB
- Data:** ChromaDB, Qdrant, Pinecone, Neo4j, Voyage

## LANGUAGES

| English | Greek  | Turkish | Arabic |
|---------|--------|---------|--------|
| C1      | Native | Native  | A2     |

## PROJECTS

**Designed an ML-powered system with XGBoost algorithm to predict employee turnover and visualize team dynamics.**

- Integrated SHAP values for model explainability, providing visual dashboards for team dynamics and risk analysis.

**Graduation Thesis: EKF-Integrated Fuzzy-CTC Control for 2-DOF Robot Manipulator**

- Developed a hybrid architecture combining CTC with Fuzzy logic to dynamically compensate for non-linear uncertainties.

**Life OS: Agentic Knowledge System**

- Architected a personal knowledge operating system utilizing Neo4j graph databases for complex semantic relationship mapping.

**Matchfluence: Influencer Marketing Marketplace**

- Engineered the scalable backend infrastructure for a B2B SaaS platform utilizing Supabase.

## CERTIFICATES

- Control Design Onramp with Simulink
- Simulink Onramp MATLAB Onramp
- IEEE - Power in Energy
- IEEE - RLC Days